# System-wide entropy from one nearest-cluster distance

Growing histories, neighborhood geometry, and a shared statistical law

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**The question.** As observations accumulate, a history acquires new coordinates. Let $u^{(n)} = (u_1, \dots, u_n)$ and $v^{(n)} = (v_1, \dots, v_n)$ be two histories of $n$ observations, where $u_t, v_t$ are their values at coordinate $t$. Writing $\|\cdot\|$ for the Euclidean norm, each new coordinate gives

$$\|u^{(n+1)} - v^{(n+1)}\|^2 = \|u^{(n)} - v^{(n)}\|^2 + (u_{n+1} - v_{n+1})^2.$$

Histories consequently separate and fixed-radius neighborhoods can empty. Can a measurement at just one cluster reveal the entropy of the larger system? The answer is yes in an identifiable statistical model: the same unknown parameter must govern both centroid separation and constituent uncertainty. The growing record then turns one distance into many observations.

**Model and observable.** Fix $K \geq 2$ clusters, each containing $m \geq 1$ labeled constituents. Use $i = 1, \dots, K$ for clusters, $\ell = 1, \dots, m$ for constituents, and $t = 1, 2, \dots$ for coordinates. Let $B_{i\ell t}$ be mutually independent binary variables, equal to 1 with common unknown probability $p \in (0, 1)$ and 0 otherwise. Constituent $\ell$ of cluster $i$ has value $X_{i\ell t} = aB_{i\ell t}$ at coordinate $t$, where $a > 0$ is known. After $n$ coordinates, define the centroid vector $C_i^{(n)}$, whose $t$th entry is $C_{it}$, and its nearest-neighbor distance $R_n$ from cluster 1:

$$C_i^{(n)} = \left( \frac{1}{m} \sum_{\ell=1}^{m} X_{i\ell t} \right)_{t=1}^{n}, \qquad R_n = \min_{2 \leq j \leq K} \|C_1^{(n)} - C_j^{(n)}\|.$$

Here $j$ indexes the other clusters. Only this distance is measured; the neighbor's identity need not be retained. Write $H_n$ for the joint Shannon entropy of all $Kmn$ constituent values, using the natural logarithm ln (entropy in nats). For a binary probability $q \in (0, 1)$, define $h_2(q) = -q \ln q - (1-q) \ln(1-q)$.

**Nearest-cluster entropy theorem.** *With $K, m, a$ fixed as $n \to \infty$, the normalized squared distance has limit $r$ below; $\xrightarrow{\text{a.s.}}$ denotes convergence with probability one:*

$$\frac{R_n^2}{n} \xrightarrow{\text{a.s.}} r = \frac{2a^2}{m} p(1-p), \qquad \frac{H_n}{Kmn} = h_2\left( \frac{1 - \sqrt{1 - 2mr/a^2}}{2} \right). \tag{1}$$

*Thus the limiting normalized distance determines the entropy per constituent per coordinate of the whole system.*

*Proof.* Write $\mathbb{E}$ for expectation and Var for variance. Each centroid coordinate has mean $ap$ and variance $a^2 p(1-p)/m$. For each $j \neq 1$, independence between clusters gives

$$\mathbb{E}\big[(C_{1t} - C_{jt})^2\big] = 2 \operatorname{Var}(C_{1t}) = \frac{2a^2}{m} p(1-p) = r.$$

The squared differences are independent and identically distributed across $t$, so the strong law gives $\|C_1^{(n)} - C_j^{(n)}\|^2 / n \to r$ almost surely. The minimum of finitely many such sequences has the same limit. Meanwhile independence of all constituents gives $H_n = Kmnh_2(p)$. Solving for $p$ leaves the two possibilities $p$ and $1 - p$, whose entropies are equal, yielding (1). □

**An estimator from a finite record.** Define $\widehat{r}_n$ as the clipped distance estimate and $\widehat{h}_n$ as the estimated entropy per constituent coordinate:

$$\widehat{r}_n = \min\left\{\frac{a^2}{2m}, \frac{R_n^2}{n}\right\}, \qquad \widehat{h}_n = h_2\left(\frac{1-\sqrt{1-2m\widehat{r}_n/a^2}}{2}\right). \tag{2}$$

Use $h_2(0) = h_2(1) = 0$ by continuity. Then $\widehat{h}_n \to h_2(p)$ almost surely. If the counts are known, $\widehat{H}_n = Kmn\widehat{h}_n$ estimates the total entropy, with

$$\frac{\widehat{H}_n - H_n}{Kmn} \xrightarrow{\text{a.s.}} 0.$$

This is consistency per constituent coordinate; it does not assert that the absolute error in an increasingly large total vanishes. For example, with $a = 1$ and $m = 10$, a limiting value $r = 0.05$ implies $p = 1/2$ and entropy $\ln 2$ per constituent coordinate. A value $r = 0.018$ implies $p = 0.1$ or $0.9$, both giving entropy approximately $0.3251$ nats.

For a finite-record guarantee, write $\mathbb{P}$ for probability and let $\varepsilon > 0$ be an error tolerance in normalized squared distance. Each squared coordinate difference lies in $[0, a^2]$; Hoeffding's inequality and a union bound over the $K - 1$ candidates give

$$\mathbb{P}\left(\left|\frac{R_n^2}{n} - r\right| > \varepsilon\right) \le 2(K-1)\exp\left(-\frac{2n\varepsilon^2}{a^4}\right). \tag{3}$$

Candidate distances share the reference cluster, but independence between different candidates is unnecessary for this union bound.

**What the neighborhood contributes.** Equation (1) also gives $R_n/\sqrt{n} \to \sqrt{r} > 0$ almost surely. Every fixed-radius reference neighborhood therefore eventually loses its last other centroid. The radius needed to retain a neighbor grows as $\sqrt{nr}$. With $a$ and $m$ fixed, its coefficient determines the entropy density through the binary model. The nearest neighbor is useful because even the last surviving neighbor carries the shared statistical signal. Here the substantive information is in the calibrated growth coefficient, rather than in the bare fact that distances grow.

**Why the shared law is essential.** Place centroid $i$ at $c_i^{(n)} = (i, \ldots, i)$ and give each cluster two constituents at equal and opposite coordinate offsets. In system A, every offset magnitude is deterministically $1/4$. In system B, independently for each cluster and coordinate, the magnitude is $1/4$ or $1/3$ with equal probability. Both systems have identical centroids and nearest distances $R_n = \sqrt{n}$. Yet their constituent entropies are respectively $0$ and $Kn\ln 2$. Centroid geometry alone cannot reveal arbitrary internal entropy. Known scale $a$, known cluster size $m$, a common binary law, independent coordinates, and the fixed number of clusters make the theorem identifiable.

**Thermodynamic meaning and scope.** If the binary values encode chamber occupancy, multiplying Shannon entropy by Boltzmann's constant $k_\mathrm{B}$ gives the coarse-grained mixing entropy. If coordinates instead record successive independent observations, $H_n$ is entropy of the accumulated record. It can grow at equilibrium while the entropy of the instantaneous physical state stays constant. The result establishes inference of system-wide uncertainty from a local distance measurement; a derivation of thermodynamic irreversibility would require additional dynamical information.

*Project source:* `expanding-neighborhoods-transcript.md`, "A theorem for your coordinate-growth model." The concentration estimate above makes the finite-record implication explicit.